

Functional And Somatometric Profiling Of Soccer Super League Vs National Team Female Players

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Soccer is a very demanding field sport characterized by multidimensional performance features.

PURPOSE: This study compared somatometric and performance characteristics of the women's Greek Super League (SL) and the National (N) team players.

METHODS: Eighteen female soccer players participated in this study. Nine of them (age: 22.6 ± 4.9 yrs; soccer experience: 7.7 ± 1.3 yrs) were members of the national (N) team and nine were members of the Super League (SL) team (age: 21.9 ± 2.4 yrs; soccer experience: 10.0 ± 4.4 yrs). Anthropometric and functional measurements were performed after completing the preparation period for the National Women's Championship. Student T-test was used for statistics and data are presented as mean $\pm$ SD.

RESULTS: No differences in the age or soccer experience were found between the two teams. Body mass index (SL: 22.3 ± 1.2 vs N: 22.3 ± 2.4) and body fat percentage (%) (SL: 15.1 ± 2.6 vs N: 14.9 ± 4.2) did not differ between the teams ($p > 0.05$). Resting heart rate was not different between two groups (89.0 ± 14.5 bpm vs 76.1 ± 20.0 bpm; $p > 0.05$). However, peak heart rate (180.1 ± 16.9 bpm vs 151.2 ± 34.0 ; $p < 0.05$) and estimated maximal oxygen uptake ($\text{VO}_{2\text{max}}$: 44.17 ± 3.29 ml/kg/min vs 38.94 ± 2.49 ml/kg/min; $p < 0.05$) achieved during the 20 m shuttle run test were significantly higher in the N team players compared to the SL players. Agility T-Test (SL: 11.1 ± 0.5 sec vs N: 11.0 ± 0.2 sec), hand grip strength of the dominant (SL: 31.2 ± 5.0 Kg vs N: 29.2 ± 1.9 Kg) and the non-dominant hand (SL: 29.1 ± 1.6 Kg vs N: 25.3 ± 5.2 Kg), counter-movement jump (SL: 23.8 ± 2.0 cm vs N: 24.0 ± 2.1 cm), and squat jump (SL: 22.5 ± 2.6 cm vs N: 24.0 ± 2.0 cm) showed no significant differences between the two teams ($p > 0.05$).

CONCLUSIONS: This study compared physical performance characteristics of soccer players of two elite women's soccer teams, revealing that aerobic capacity may be a distinguishing variable between those high-level teams and a potential criterion parameter for player selection in those teams.

Evaluation Of Player Load In Division I Women's Soccer

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Sport coaches use athlete monitoring systems to gauge workloads and intensity during training and games. In Playertek devices, Player Load (PL) provides a measure of training volumes in arbitrary units (AU).

PURPOSE: The purpose of this study was to evaluate a proprietary metric from Playertek, PL, for collegiate women's soccer across different positions, and compare training to games.

METHODS: Global positioning units were worn by athletes ($n = 20$) during 42 training sessions and 15 games. Data were uploaded to Playertek Software and condensed to eliminate downtime during the recorded sessions.

RESULTS: There was a difference ($p = .001$) between training PL (193.6 ± 10.4 AU) and game PL (399.9 ± 99.1 AU). However, there were no differences by position for training PL ($p = .347$) or game PL ($p = .425$) between forwards (training: 189.4 ± 13.1 AU; game: 389.0 ± 100.4 AU), midfielders (training: 189.4 ± 6.5 AU; game: 469.8 ± 84.9 AU), and defenders (training: 196.5 ± 9.8 AU; game: 387.0 ± 102.1 AU).

CONCLUSIONS: The data suggests a difference between training and games. This finding may be due to differences in playing time and distribution of positions within the sample. Another possible explanation for the difference in PL is that the time and structure of training may lead to more downtime than during games. Further research should adjust for time and intensity for training versus games. This information would allow coaches to better match games demands during training.

Investigating The Effects Of 3-Dimensional Motion Object Tracking (3D-MOT) Training On In-game College Soccer Performance

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BACKGROUND: Visual training has previously been shown to correlate to sport specific level of training and sport specific performance measures in controlled conditions. However, it remains unclear if these relationships exist between 3D-MOT training and in competition decision making metrics over the duration of a soccer season. The purpose of this study was to investigate the relationship between the effects of 4-weeks of 3D-MOT training on in-game soccer performance measures.

METHODS: 25 NCAA Division I soccer players were split into two groups, after exclusionary criteria (played > 10 minutes per game) only 13 were utilized for analysis NT ($n = 6$) and CON ($n = 7$). Participants completed a baseline assessment of visual tracking speed (VTS) and subsequently 10 training sessions composed of 20 trials each were performed by the Neurotracker (NT) group over a 4-week period on a 3D-MOT software Neurotracker (NT; CogniSens Athletic, Inc., Montreal, Quebec, Canada). The soccer performance metrics were obtained from WyScout where 2 game averages were examined to represent pre-NT and post-NT (Wyscout, Chiavari, Italy). An analysis of covariance (ANCOVA) was utilized to compare NT and CON mean differences among variables.

RESULTS: There was a no significant difference between the CON group ($53.96 \pm 9.55\%$) and NT group ($58.38 \pm 11.98\%$) successful actions post-NT ($p = 0.453$, $\eta^2 = 0.057$). There was a no significant difference between CON group ($64.79 \pm 9.97\%$) and NT group ($71.71 \pm 12.35\%$) passing accuracy post-NT ($p = 0.304$, $\eta^2 = .105$). There was a no significant difference between the CON group ($76.1 \pm 3.1\%$) and NT group ($77.0 \pm 3.3\%$) Short+medium passes post-NT ($p = 0.847$, $\eta^2 = 0.004$).

CONCLUSIONS: This is the first study to examine the effects of Neurotracker on collegiate in-game soccer performance metrics related to in-game decision-making. While none of the metrics of interest were found to have a significant difference, passing accuracy was found to have a moderate effect ($\eta^2 = .105$) with the NT group increasing passing accuracy by 8.5% (pre: 63.2% vs post: 71.7%) compared to the CON group increasing by 3.5% (pre: 61.3% vs post: 64.8). Further research is needed utilizing multiple sites to increase sample size and examine a potential transfer effect through training with 3D-MOT given the practical significance of improved passing accuracy.

F-22 Vascular Function: Brain

Baseline Hypocapnia Blunts Steady-state But Not Transient Hypercapnia-induced Shear-mediated Dilation Of The Internal Carotid Artery

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PURPOSE: Shear-mediated dilation of the internal carotid artery (ICA) induced by transient or steady-state hypercapnia (TR and SS tests, respectively) potentially reflects cerebrovascular endothelial function. During the baseline phase prior to elevating $P_{\text{ET}}\text{CO}_2$, increased ventilation and/or hypocapnia associated with hyperventilation may attenuate the shear rate (SR), resulting in suppression of ICA dilation. This study aimed to elucidate whether hyperventilation with or without hypocapnia blunts shear-mediated dilation of the ICA and whether this effect differs between the TR and SS tests.